

III Year

Roll No.

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Course Name: Agricultural Microbiology

Course Code: 304

Branch(s) Microbiology

Time: 120 Min

Max. Marks: 60

Instructions:

1. Attempt all the sections.
2. Assume missing data suitably, if any.

CO1: Introduction and historical background of soil microbiology soil microflora- bacteria, fungi, actinomycetes, algae, protozoa, and viruses, Rhizosphere, Rhizoplane.

CO-2: Role of microbes in weathering of minerals and soil formation, components of soil, texture and soil profile.

KL: Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create

Section-A

Attempt all questions of this section.

(2 x 6 = 12 Marks)

Q.No.	Questions	CO	KL
1.	a Define rhizoplane.	1	K1
	b What are actinomycetes?	1	K1
	c Define soil microbiology.	1	K2
	d How do microorganisms contribute to the weathering of minerals?	2	K1
	e What is the role of microbes in soil formation.	2	K2
	f Define rhizosphere.	2	K3

Section-B

Attempt any two questions of this section.

(7 x 2 = 14 Marks)

Q.No.	Questions	CO	KL
2.	Explain the major groups of soil microflora with suitable examples, including bacteria, fungi and actinomycetes.	1	K1
3.	Explain the concept of rhizosphere and rhizoplane. Discuss their importance in plant-microbe interactions.	1	K1
4.	Discuss the ecological roles of algae, protozoa and viruses in soil.	1	K1

Section-C

Attempt any two questions of this section.

(7 x 2 = 14 Marks)

5.	Explain the role of microorganisms in the weathering of minerals and their contribution to soil formation.	2	K3
6.	Describe the historical development and scope of soil microbiology.	2	K2
7.	Describe the significance of the rhizoplane as a site of microbial colonization and plant-microbe interactions.	2	K2

Section-D**Attempt all questions of this section.****(10 x 2 = 20 Marks)**

Q.No.	Questions	CO	KL
8.	Explain how microbial interactions in the rhizosphere can influence plant growth and soil fertility.	1	K2
	Or		
	Describe the physical, chemical and biological components of soil and explain their significance.	1	K2
9.	A soil sample shows a high population of bacteria and fungi near plant roots. Explain the possible reasons for this observation.	2	K3
	Or		
	How can changes in soil environmental conditions affect the distribution of different groups of soil microflora?	2	K3